Harivijay R Gunaseelan

+1 (226) 752-7014 | Email: hrathina@uwaterloo.ca | Profile: https://28harivijay.github.io/
LinkedIn: linkedin.com/in/harivijay-rg | Github: github.com/28harivijay

Summary

Computer Engineering student at the University of Waterloo with hands-on experience in **embedded systems, firmware development, digital design, and hardware-software integration**. Experienced developing embedded applications using **ESP32, Arduino, C++, Verilog, and FPGA platforms**. Passionate about designing reliable, efficient firmware and embedded solutions for real-world systems, with a strong foundation in low-level programming and hardware debugging.

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science (BASc), Computer Engineering | 2025 - Present

- **Coursework:** Digital Computers, Numerical Methods, C++ (DSA, OOP), Embedded Systems (Arduino, sensors)

Skills

Languages: Python, C++, Embedded C++ (Arduino), MATLAB, JavaScript, C#, HTML, CSS
AI/ML Tools: LangChain, FAISS, HuggingFace Embeddings, Scikit-learn, Pandas, RAG Pipelines, Groq API
Backend & APIs: FastAPI, Django, REST APIs, Webhooks, CORS, Cloudflare Tunnel
Hardware & Embedded: ESP32, I2C , Arduino Nano, Biomedical Sensors (SpO_2, Accelerometer)
Tools: Git, GitHub, VS Code, Render, Vercel, React, Arduino IDE, Unity, Bootstrap, CI/CD pipelines

Work Experience

Accessibility Review Team Member — University of Waterloo
Waterloo, ON (Remote, Project-Based) June 2026 - August 2026

- Audited **HTML source code** across **3 UW PD courses** against **WCAG accessibility standards**, identifying and documenting content quality issues impacting assistive technology users across multiple course modules.
- Produced **structured compliance reports** with cited WCAG criteria for each finding, delivering clear, verifiable, and actionable documentation that enabled stakeholders to resolve accessibility gaps efficiently.
- Aligned classification and validation standards through recurring team syncs and created structured, screen-reader-friendly transcripts translating visual course content into accessible text.

Projects

PRISM — Patient Response Insight & Surveillance Monitor
View Repository | *Hardware Lead* | View in Portfolio Nov 2025 - Dec 2025

- Developed a wearable **HID (Hospital Induced Delirium) detection system** using **ESP32** and biomedical sensors, reducing nursing monitoring workload by **20%** and enabling significantly faster clinical response times.
- Programmed **real-time embedded firmware in C++ (Arduino)** for continuous biomedical data acquisition, optimizing buffering and I2C communication handling to minimize sensor data loss under live operating conditions.
- Benchmarked **ESP32** against Arduino Nano R4 Minima across key data throughput metrics, achieving **15× faster** handling for SpO_2 and accelerometer sensor streams in a resource-constrained wearable environment.

Research Presentation — Neuromorphic Computing for Edge AI | View in Portfolio Oct 2025 - Nov 2025

- Investigated **neuromorphic computing architectures** as a solution to energy inefficiency in edge AI inference, analyzing how **spiking neural networks (SNNs)** reduce power consumption by up to **10–100×** over conventional deep learning models.
- Evaluated **STDP** vs **surrogate-gradient** learning methods across **5+ peer-reviewed studies**, comparing convergence speed and inference accuracy, and communicated findings through a **technical poster** on event-driven neural computation.

AI GitHub PR (Pull Request) Review Bot
View Repository | *Personal Project* | View in Portfolio April 2026 - May 2026

- Built an **AI-driven PR review bot** using **FastAPI** and **LangChain**, integrating **GitHub Webhooks** to automatically trigger real-time code reviews on every new pull request without any manual intervention required.
- Fetched PR diffs via **GitHub REST API** and constructed **structured LangChain prompt chains** to route code changes through **Groq (LLaMA 3.3)**, generating categorized feedback on bugs, readability, and potential improvements per commit.
- Exposed the **FastAPI webhook server** to GitHub's event delivery system using a **Cloudflare Tunnel**, enabling fully automated end-to-end PR reviews to complete within **10–15 seconds** of a pull request being opened.

Achievements & Certifications

- **FreeCodeCamp Python Developer Certification**: 300-hour certification in Python fundamentals, OOP & DSA.
- **MATLAB Onramp** (MathWorks): Certification in MATLAB syntax, matrix operations, data import and plotting.